```
[s]@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:27:d8:cc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86069sec preferred_lft 86069sec
    inet6 fe80::a00:27ff:fe27:d8cc/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[s]@localhost ~]$
```

```
[s]@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[s]@localhost ~]$
```

```
[s]@localhost ~]$ hostname
localhost.localdomain
[s]@localhost ~]$
```

```
[s]@localhost ~]$ ping google.com
PING google.com (142.251.43.142) 56(84) bytes of data.
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=1 ttl=118 time=6.27 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=2 ttl=118 time=7.93 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=3 ttl=118 time=11.1 ms
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=4 ttl=118 time=5.11 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=5 ttl=118 time=12.3 ms
```

```
[s]@localhost ~]$ nslookup google.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.205.110
Name:   google.com
Address: 2404:6800:4007:815::200e

[s]@localhost ~]$
```

```
[s]@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
udp6       0      0 localhost:323           [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
```

```
[s]@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[s]@localhost ~]$
```

```
[s]@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether 08:00:27:27:d8:cc brd ff:ff:ff:ff:ff:ff
[s]@localhost ~]$
```

```
[s]@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether 08:00:27:27:d8:cc brd ff:ff:ff:ff:ff:ff
[s]@localhost ~]$
[s]@localhost ~]$ ss -tuln
Netid       State      Recv-Q      Send-Q            Local Address:Port             Peer Address:Port
udp         ESTAB      0           0           10.0.2.15%enp0s3:68              10.0.2.2:67
[s]@localhost ~]$
```

```
[s]@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

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Exp No: 1

Date: 16/7/26

Basic Networking Commands in Linux and Windows

Aim: To study and execute basic networking commands available in Linux and Windows operating systems for diagnosing and troubleshooting network issues.

Requirements:- Pc with windows/Linux os, Terminal or command prompt, Active network connection

Theory/Background:-

A computer network is a collection of interconnected devices that share resources and information. Networking commands are built-in utilities that allow administrators and users to configure, diagnose, and troubleshoot network connection.

Windows provides commands such as ipconfig, ping, tracert, netstat, nslookup, arp, and route, Linux provides equivalents: ifconfig/ip, ping, traceroute, netstat/ss, nslookup/dig, arp, and route/ip route.

These commands operate at various layers of the osi model - Ping works at the network layer (ICMP), netstat at the transport layer, and ipconfig/ifconfig at the Data Link/ network layers.

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Windows Commands

Commands		Discription	Sample o/p
ipconfig	-	IP Address	192.168.1.20
ipconfig /all	-	MAC Address, DHCP server, DNS server	Windows IP configuration,
ping google.com	-	checks destination is reachble	sent = 4 Received = 4 Lost = 0
Tracet google.com	-	displays every router between computer and..	192.168.1.1 ISP Router nogle Network
nslookup google.com	-	DNS server. IP address of website	Name: Address:
route print	-	Network Routes Gateway	Interface List
netstat	-	active TCP, Litening ports,	Active connectio TCP UBP
netstat -a	-	All ports TCP connections UPP ports	Active connectio TCP UBP
getmac	-	MAC address	pA TN
hostname	-	computer name	DESKTOP-ABC123
arp -a	-	IP to MAC Address	Interface address.

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Procedure:-

1. Open command prompt (Windows) or terminal.
2. Execute "ip config" (windows) or 'ifconfig'/ip a' (Linux) to view IP configuration.
3. Use 'ping 8.8.8.8' to test connectivity to Google's DNS server.
4. Execute 'tracert' google.com' (windows) or 'traceroute' google.com' (Linux) to view Packet Path.
5. Run 'netstat' -an' to list active network connections and listening ports.
6. Use 'nslookup google.com' to perform DNS lookup.
7. Execute 'arp -a' to view the ARP cache.
8. Use 'route print' (windows) or 'route -n' (Linux) to view the routing table.
9. Record all outputs and analyze the result.

Result:-

Basic networking commands were successfully executed on both Linux and windows IP address, routing tables, ARP caches, and DNS resolution were observed and recorded.

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